

# Multiple-Input Transfer Function Analysis for Cerebrovascular Regulation

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# Presentation Roadmap

1. Physiological motivation: cerebrovascular autoregulation
2. Transfer function analysis and spectral estimation
3. Coherence as a reliability measure
4. CARNet recommendations for standard TFA
5. Motivation for multiple-input modeling
6. MISO transfer function formulation
7. Simulation pipeline and SISO vs. MISO comparison
8. Coherence-based interpretation of MISO results
9. Next steps: real data and wavelet-based MISO

# What is Cerebrovascular Autoregulation?

- ▶ The brain's ability to maintain relatively stable cerebral blood flow despite changes in systemic blood pressure.
- ▶ Healthy cerebral vessels can constrict or dilate to buffer pressure changes and preserve brain perfusion.
- ▶ Impaired autoregulation can occur in brain injury or neurological disease, making it clinically important to quantify.

# What is a Transfer Function?

- ▶ A transfer function describes how an input signal is transformed into an output signal.
- ▶ In the frequency domain, each input frequency component is scaled and phase-shifted:

$$Y(f) = H(f)X(f)$$

- ▶ The transfer function  $H(f)$  contains:
  - ▶ **Gain:** how strongly the output responds at frequency  $f$
  - ▶ **Phase:** how shifted or delayed the output is at frequency  $f$

# Applying TFA to Autoregulation: SISO MAP $\rightarrow$ CBV

- ▶ We first think of cerebral blood flow velocity,  $cbv(t)$ , as a function of mean arterial pressure,  $map(t)$ .
- ▶ In the time domain, this relationship is represented as a convolution between the input signal and the impulse response:

$$cbv(t) = (map * h_{map \rightarrow cbv})(t) = \int_{-\infty}^{\infty} map(\tau) h_{map \rightarrow cbv}(t - \tau) d\tau$$

- ▶ In the frequency domain, convolution becomes multiplication:

$$CBV(f) = H_{map \rightarrow cbv}(f) MAP(f)$$

- ▶ Therefore, the transfer function can be written as:

$$H_{map \rightarrow cbv}(f) = \frac{CBV(f)}{MAP(f)}$$

# Shortcoming of Direct FFT Division

- ▶ From the basic frequency-domain relationship, a simple estimate would be:

$$H_{map \rightarrow cbv}(f) = \frac{CBV(f)}{MAP(f)}$$

- ▶ However, this estimate can become unstable when:
  - ▶  $MAP(f)$  is very small at a frequency
  - ▶ the input or output contains noise
  - ▶ a single FFT estimate is dominated by random variation
- ▶ Therefore, direct FFT division can produce unreliable gain and phase estimates.

# Estimating the Transfer Function with Power Spectra

- ▶ Instead of dividing single FFTs, we estimate the relationship using averaged spectral quantities.
- ▶ Start with the frequency-domain model:

$$CBV(f) = H_{map \rightarrow cbv}(f)MAP(f) + \epsilon(f)$$

- ▶  $\epsilon(f)$  represents output variation not explained by  $map(t)$ .
- ▶ Multiply both sides by the complex conjugate of the input,  $MAP^*(f)$ :

$$MAP^*(f)CBV(f) = H_{map \rightarrow cbv}(f)MAP^*(f)MAP(f) + MAP^*(f)\epsilon(f)$$

# From Spectral Averaging to the Transfer Function

- ▶ Define the cross-spectrum and auto-spectrum as:

$$S_{map,cbv}(f) = E [MAP^*(f)CBV(f)]$$

$$S_{map,map}(f) = E [MAP^*(f)MAP(f)]$$

- ▶ If unexplained output variation is gaussian in nature and independent to  $map(t)$ :

$$E [MAP^*(f)\epsilon(f)] \approx 0$$

$$S_{map,cbv}(f) = H_{map \rightarrow cbv}(f)S_{map,map}(f)$$

$$H_{map \rightarrow cbv}(f) = \frac{S_{map,cbv}(f)}{S_{map,map}(f)}$$



# Welch's Method for Finding PSDs

- ▶ Power spectral density estimates how much signal power exists at each frequency.
- ▶ A single FFT can be noisy, so Welch's method improves the estimate by averaging across smaller time windows.
- ▶ General workflow:
  1. Split the signal into overlapping windows
  2. Apply a window function to each segment
  3. Compute the FFT of each windowed segment
  4. Average the spectral estimates across windows

Single FFT estimate     $\longrightarrow$     averaged spectral estimate

# Welch's Method for the Auto-Spectrum

- ▶ Goal: estimate how much power  $map(t)$  has at each frequency.
- ▶ Start with the time-domain signal:

$$map(t)$$

- ▶ Split the signal into  $M$  shorter, overlapping segments:

$$map_1(t), map_2(t), \dots, map_M(t)$$

- ▶ Apply a window function such as hann to each segment:

$$\widetilde{map}_m(t) = w(t)map_m(t)$$

- ▶ Compute the FFT of each windowed segment:

$$MAP_m(f) = \mathcal{F} \{ \widetilde{map}_m(t) \}$$

# Auto-Spectrum: Averaging Power Across Windows

- ▶ For each window, estimate power by multiplying the frequency-domain signal by its complex conjugate:

$$MAP_m^*(f)MAP_m(f) = |MAP_m(f)|^2$$

- ▶ Then average the power estimates across all  $M$  windows:

$$S_{map,map}(f) = \frac{1}{M} \sum_{m=1}^M MAP_m^*(f)MAP_m(f) = \frac{1}{M} \sum_{m=1}^M |MAP_m(f)|^2$$

- ▶ This gives the auto-spectrum of  $map(t)$ .
- ▶  $S_{map,map}(f)$  tells us the MAP power available at each frequency.

# Cross-Spectrum: Shared Frequency Content

- ▶ The cross-spectrum measures how two signals vary together at each frequency.
- ▶ For  $map(t)$  and  $cbv(t)$ , Welch's method averages the input-output product across windows.

$$S_{map,cbv}(f) = \frac{1}{M} \sum_{m=1}^M MAP_m^*(f) CBV_m(f)$$

- ▶ Unlike the auto-spectrum, the cross-spectrum is complex.
- ▶ Its magnitude captures shared frequency content.
- ▶ Its angle captures relative phase between  $map(t)$  and  $cbv(t)$ .

# Coherence: Reliability of SISO TFA

- ▶ Using the spectra we can solve for the transfer function to estimate gain and phase at each frequency.
- ▶ However, gain and phase are only meaningful if the input-output relationship is reliable at that frequency.
- ▶ Coherence measures how strongly an input and output are linearly related at each frequency.
- ▶ For the MAP-to-CBV SISO relationship:

$$\gamma_{map,cbv}^2(f) = \frac{|S_{map,cbv}(f)|^2}{S_{map,map}(f)S_{cbv,cbv}(f)}$$

- ▶ Coherence ranges from 0 to 1.
- ▶ High coherence means gain and phase are more reliable at that frequency.
- ▶ Low coherence means the transfer function estimate should be interpreted cautiously or excluded.

# CARNet Recommendations: Data Acquisition

- ▶ Recording duration:

$$T_{\text{recording}} \geq 5 \text{ min}$$

- ▶ Sampling frequency:

$$f_s \geq 50 \text{ Hz}$$

- ▶ Anti-aliasing low-pass filter:

$$f_c = 20 \text{ Hz}$$

# Pre-Processing: Beat-to-Beat Averaging

- ▶ Start with raw recordings:

$$map_{raw}(t), \quad cbv_{raw}(t)$$

- ▶ Compute discrete beat-to-beat means for each cardiac cycle.

$$map_i = \frac{1}{N_i} \sum_{n=n_i}^{n_{i+1}-1} map_{raw}[n]$$

$$cbv_i = \frac{1}{N_i} \sum_{n=n_i}^{n_{i+1}-1} cbv_{raw}[n]$$

- ▶ The times  $n_i$  and  $n_{i+1}$  are found using systolic peaks in the raw pressure waveform, or using R-peaks if ECG is recorded.

# Pre-Processing: Why Beat-to-Beat?

- ▶ The spacing between beats is not constant.
- ▶ One cardiac cycle may last 0.7 s while another lasts 0.8 s.
- ▶ Therefore, the beat-to-beat sequence exists at uneven time points:

$$(t_1, map_1), (t_2, map_2), (t_3, map_3), \dots$$

- ▶ Raw waveforms are not used directly because TFA for dCA focuses on slow MAP–CBV fluctuations, not every detail of the waveform.



# Pre-Processing: Resampling and Cleaning

- ▶ Frequency analysis requires evenly spaced samples.
- ▶ CARNet recommends interpolating the beat-to-beat sequence onto an evenly spaced time grid with at least a 4 Hz resampling rate:

$$t = 0, 0.25, 0.50, 0.75, 1.00, \dots$$

- ▶ Visually inspect beat-to-beat data.
- ▶ Replace short artifacts up to about 3 beats with linear interpolation.
- ▶ Remove mean values:

$$x_{\text{centered}}(t) = x(t) - \bar{x}$$

- ▶ Avoid detrending prior to TFA.
- ▶ Avoid low-pass or high-pass filtering prior to TFA.

# Pre-Processing Summary

- ▶ The overall preprocessing pipeline is:

$$map_{raw}(t) \rightarrow map_i \rightarrow map_{interp}(t) \rightarrow map_{clean}(t) \rightarrow map_{centered}(t)$$

$$cbv_{raw}(t) \rightarrow cbv_i \rightarrow cbv_{interp}(t) \rightarrow cbv_{clean}(t) \rightarrow cbv_{centered}(t)$$

- ▶ These centered, evenly sampled signals are then used for spectral analysis.

# Welch/TFA Settings: Windowing

- ▶ Window type: Hann

- ▶ Window length:

$$T_{\text{window}} = 100 \text{ s}$$

- ▶ Frequency resolution:

$$\Delta f = \frac{1}{T_{\text{window}}} = \frac{1}{100} = 0.01 \text{ Hz}$$

- ▶ Window overlap: 50% overlap

$$N_{\text{window}} = 100f_s$$

$$N_{\text{overlap}} = 0.5N_{\text{window}}$$

# Welch/TFA Settings: Spectral Smoothing

- ▶ After computing the spectra, smooth both auto-spectra and cross-spectra with a triangular window:

$$\left[ \frac{1}{4}, \frac{1}{2}, \frac{1}{4} \right]$$

- ▶ At each frequency bin  $k$ , this means:

$$S_{\text{smooth}}[k] = \frac{1}{4}S[k-1] + \frac{1}{2}S[k] + \frac{1}{4}S[k+1]$$

- ▶ This smoothing is applied to both the auto-spectra and cross-spectra before estimating gain, phase, and coherence.

# Coherence Thresholding

Coherence is defined as:

$$\gamma_{map,cbv}^2(f) = \frac{|S_{map,cbv}(f)|^2}{S_{map,map}(f)S_{cbv,cbv}(f)}$$

- ▶ Coherence ranges from 0 to 1.
- ▶ Higher coherence means the MAP–CBV relationship is more reliable at that frequency.
- ▶ CARNet recommends excluding gain and phase values in frequency bands where coherence does not exceed the critical coherence value.

| Windows | 95% Crit. Coh. |
|---------|----------------|
| 3       | 0.51           |
| 4       | 0.41           |
| 5       | 0.34           |
| 6       | 0.29           |
| 7       | 0.25           |
| 8       | 0.22           |
| 9       | 0.20           |
| 10      | 0.18           |
| 15      | 0.12           |
| 20      | 0.09           |
| 25      | 0.08           |

Critical coherence decreases as the number of averaged Welch windows increases.

# Frequency Bands of Interest

$$0.02 \text{ Hz} \leq f \leq 0.5 \text{ Hz}$$

- ▶ Report average phase and gain within these ranges:

| Band | Frequency range |
|------|-----------------|
| VLF  | 0.02–0.07 Hz    |
| LF   | 0.07–0.20 Hz    |
| HF   | 0.20–0.50 Hz    |

# Reporting Gain: Absolute Gain

- ▶ Report both absolute gain and normalized gain.
- ▶ Absolute gain uses the original units of the signals.

$$map(t) : \text{mmHg}$$

$$cbv(t) : \text{cm/s}$$

- ▶ Therefore, the units of  $H_{map \rightarrow cbv}(f)$  are:

$$\frac{\text{cm/s}}{\text{mmHg}}$$

# Reporting Gain: Normalized Gain

- Convert raw  $cbv(t)$  data into a percentage before estimating gain:

$$cbv_{\%}(t) = \frac{cbv_{raw}(t)}{cbv_{raw}} \times 100$$

- Then remove the mean from  $cbv_{\%}(t)$  before TFA.
- Since  $cbv_{\%}(t)$  has units of percent, the normalized gain has units:

$$H_{map \rightarrow cbv_{\%}}(f) = \frac{CBV_{\%}(f)}{MAP(f)}$$
$$\frac{\%}{\text{mmHg}}$$

- This makes it easier to compare gain across different participants.



# Peng et al.: Motivation for Multiple Inputs

- ▶ Traditional TFA often models cerebral blood flow velocity as a single-input system:

$$map(t) \rightarrow cbv(t)$$

- ▶ However, Peng et al. argued that  $cbv(t)$  is not determined by pressure alone.
- ▶ Other physiological variables, especially  $CO_2$ , can also affect cerebral blood flow velocity.
- ▶ Therefore, low univariate coherence between  $map(t)$  and  $cbv(t)$  does not necessarily mean the system is nonlinear.
- ▶ It may mean that important inputs are missing from the model.

# Peng et al.: Multiple-Input Model

- ▶ Peng et al. modeled CBFV as the output of multiple physiological inputs:

$$CBV(f) = H_{map \rightarrow cbv}(f)MAP(f) + H_{co2 \rightarrow cbv}(f)CO2(f) + H_{o2 \rightarrow cbv}(f)O2(f)$$

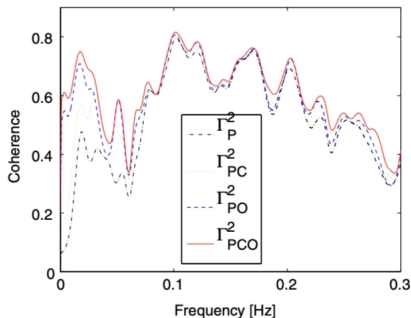
- ▶ This separates the estimated contribution of each input pathway:
  - ▶ pressure-to-flow pathway
  - ▶ CO<sub>2</sub>-to-flow pathway
  - ▶ O<sub>2</sub>-to-flow pathway
- ▶ Multiple coherence then asks how well all inputs together explain the output.

single input: map  $\rightarrow$  cbv

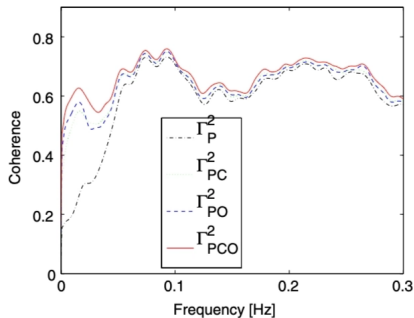
multiple inputs: map, co<sub>2</sub>, o<sub>2</sub>  $\rightarrow$  cbv

# Peng et al.: Multiple Coherence Result

- ▶ Peng et al. showed that adding PETCO<sub>2</sub> and/or PETO<sub>2</sub> increased coherence at low frequencies compared with ABP alone.
- ▶ This supports the idea that low univariate coherence may reflect missing physiological inputs rather than only nonlinearity.



(b) typical subject



(c) group averaged values

Black = ABP-only univariate coherence; colored curves = multiple coherence with added PETCO<sub>2</sub>, PETO<sub>2</sub>, or both.

- ▶ Peng et al. found that adding O<sub>2</sub> did not significantly improve coherence, likely because O<sub>2</sub> was highly correlated with CO<sub>2</sub>.

# Why Different Coherence Measures Are Needed

- ▶ Transfer function gain and phase should only be interpreted at frequencies where the estimate is reliable.
- ▶ In this project, different coherence measures answer different reliability questions.

| Measure               | Question it answers                                                         |
|-----------------------|-----------------------------------------------------------------------------|
| Univariate coherence  | Does one input alone explain $cbv(t)$ ?                                     |
| Multiple coherence    | Do $map(t)$ and $co2(t)$ together explain $cbv(t)$ ?                        |
| Partial coherence     | Does one input still explain $cbv(t)$ after accounting for the other input? |
| Input-input coherence | Are $map(t)$ and $co2(t)$ too overlapping to separate cleanly?              |

# Univariate and Multiple Coherence

- **Univariate coherence** is ordinary two-signal coherence between one input and the output.

$$\gamma_{map,cbv}^2(f) = \frac{|S_{map,cbv}(f)|^2}{S_{map,map}(f)S_{cbv,cbv}(f)}$$

- It is useful for SISO TFA, but it can be misleading if  $cbv(t)$  depends on multiple inputs.
- **Multiple coherence** asks whether the full two-input model explains the output:

$$map(t), co2(t) \rightarrow cbv(t)$$

$$\gamma_{cbv|map,co2}^2(f) = \frac{S_{\widehat{cbv},\widehat{cbv}}(f)}{S_{cbv,cbv}(f)}$$

- If multiple coherence is higher than univariate coherence, the added input is helping explain the output.

# Partial Coherence: Pathway-Specific Reliability

- ▶ Perreault et al. define partial coherence as the ordinary coherence after the effects of the other inputs have been removed from both the input and output of interest.
- ▶ In this project, MAP partial coherence asks whether  $map(t)$  still explains  $cbv(t)$  after accounting for  $co2(t)$ :

$$map(t) \rightarrow cbv(t) \quad \text{after accounting for } co2(t)$$

- ▶ This is useful because multiple coherence can be high even when the individual pathway split is uncertain.

# Partial Coherence Formula

- ▶ For MAP-to-CBV, first remove the part of the MAP-CBV relationship that can be explained through CO<sub>2</sub>:

$$S_{map,cbv|co2}(f) = S_{map,cbv}(f) - \frac{S_{map,co2}(f)S_{co2,cbv}(f)}{S_{co2,co2}(f)}$$

- ▶ Then compute coherence using the residual spectra:

$$\gamma_{map,cbv|co2}^2(f) = \frac{|S_{map,cbv|co2}(f)|^2}{S_{map,map|co2}(f)S_{cbv,cbv|co2}(f)}$$

- ▶ High partial coherence means the MAP-to-CBV pathway remains reliable after accounting for CO<sub>2</sub>.

# Input-Input Coherence: Input Overlap

- ▶ Input-input coherence is still ordinary two-signal coherence:

$$\gamma_{map,co2}^2(f) = \frac{|S_{map,co2}(f)|^2}{S_{map,map}(f)S_{co2,co2}(f)}$$

- ▶ Here, it is not measuring whether an input explains CBV.
- ▶ Instead, it measures whether  $map(t)$  and  $co2(t)$  overlap at the same frequency.
- ▶ Perreault et al. note that independent inputs have coherence near 0, perfectly linearly dependent inputs have coherence 1, and perfectly coupled inputs cause the identification procedure to fail.
- ▶ Therefore, high input-input coherence is an identifiability warning: the full MISO model may explain  $cbv(t)$ , but pathway-specific gains may be harder to separate.



# Our MISO Model

- ▶ Motivated by Peng et al., we modeled  $cbv(t)$  as the output of two simultaneous inputs:
  - ▶ mean arterial pressure,  $map(t)$
  - ▶ carbon dioxide,  $co2(t)$
- ▶ In the frequency domain:

$$CBV(f) = H_{map \rightarrow cbv}(f)MAP(f) + H_{co2 \rightarrow cbv}(f)CO2(f)$$

- ▶ The goal is to estimate both transfer functions at each frequency:

$$H(f) = \begin{bmatrix} H_{map \rightarrow cbv}(f) \\ H_{co2 \rightarrow cbv}(f) \end{bmatrix}$$

# Solving the MISO Transfer Function

- ▶ Using the same spectral logic from the SISO derivation, the MISO system becomes:

$$\underbrace{\begin{bmatrix} S_{map,map}(f) & S_{map,co2}(f) \\ S_{co2,map}(f) & S_{co2,co2}(f) \end{bmatrix}}_{\text{input overlap matrix}} \underbrace{\begin{bmatrix} H_{map \rightarrow cbv}(f) \\ H_{co2 \rightarrow cbv}(f) \end{bmatrix}}_{\text{transfer functions}} = \underbrace{\begin{bmatrix} S_{map,cbv}(f) \\ S_{co2,cbv}(f) \end{bmatrix}}_{\text{input-output relationships}}$$

- ▶ Then solve for the transfer function vector:

$$H(f) = S_{xx}^{-1}(f)S_{xy}(f)$$

# Simulation Setup: Input Signals

- ▶ The simulated inputs were designed with known frequency components in VLF, LF, and HF ranges.
- ▶ MAP and CO<sub>2</sub> also include overlapping/near-overlapping frequencies to test whether MISO can separate input contributions.

| Input    | VLF     | LF      | HF      |
|----------|---------|---------|---------|
| $map(t)$ | 0.03 Hz | 0.10 Hz | 0.30 Hz |
| $co2(t)$ | 0.05 Hz | 0.11 Hz | 0.40 Hz |

$$shared_{LF}(t) = \sin(2\pi(0.08)t)$$

$$map(t) = \sin(2\pi(0.03)t) + \sin(2\pi(0.10)t) + \sin(2\pi(0.30)t) + 0.4shared_{LF}(t)$$

$$co2(t) = \sin(2\pi(0.05)t) + \sin(2\pi(0.11)t) + \sin(2\pi(0.40)t) + 0.4shared_{LF}(t)$$

- ▶ 30 dB white Gaussian noise was added to both input signals.

# Simulation Setup: Output Signal

- ▶ The simulated output  $cbv(t)$  was generated from known MAP and CO<sub>2</sub> contributions.
- ▶ Each component was assigned a known gain and phase shift.

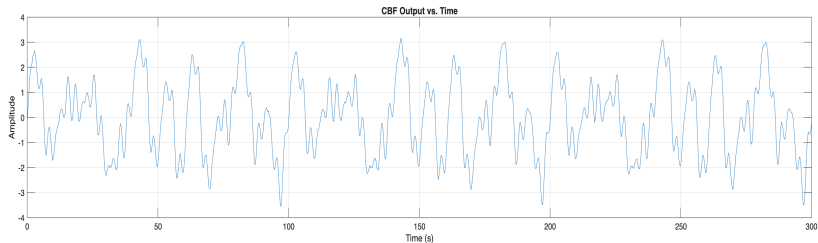
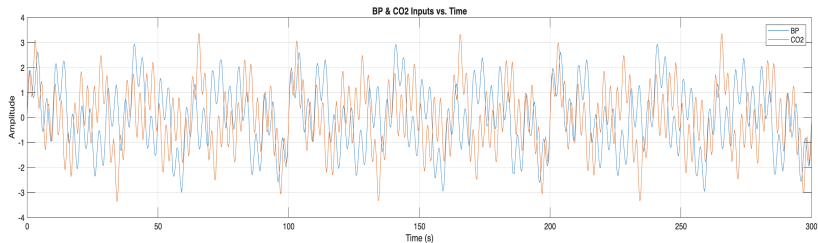
| Source              | Frequency | Gain | Phase     |
|---------------------|-----------|------|-----------|
| MAP VLF             | 0.03 Hz   | 0.7  | $-\pi/3$  |
| MAP LF              | 0.10 Hz   | 1.0  | $-\pi/6$  |
| MAP HF              | 0.30 Hz   | 0.5  | $-3\pi/4$ |
| CO <sub>2</sub> VLF | 0.05 Hz   | 1.2  | $+\pi/4$  |
| CO <sub>2</sub> LF  | 0.11 Hz   | 0.6  | $+\pi/2$  |
| CO <sub>2</sub> HF  | 0.40 Hz   | 0.4  | $-\pi/8$  |
| Shared LF           | 0.08 Hz   | 0.8  | —         |

$$cbv(t) = cbv_{MAP}(t) + cbv_{CO_2}(t) + 0.8shared_{LF}(t) + \epsilon(t)$$

- ▶ 30 dB white Gaussian noise was also added to the output.

# Simulated Time Series

- The resulting time-domain signals were used as the test case for the MISO TFA pipeline.

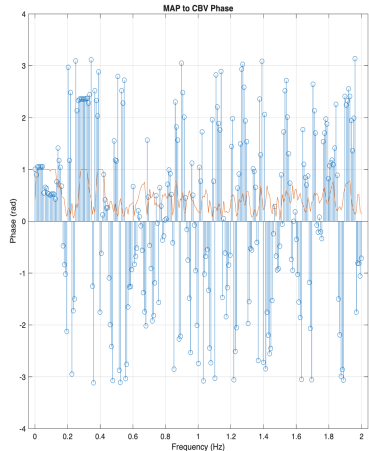
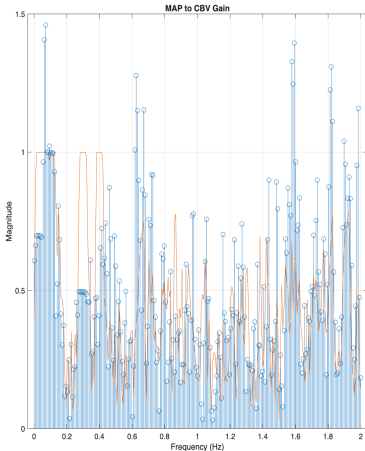


# Initial MISO Estimate: MAP to CBV

- ▶ First, we estimated the MAP-to-CBV transfer function from the MISO model before applying coherence-based filtering.
- ▶ The orange curve shows multiple coherence, overlaid to indicate which frequencies have more reliable MISO estimates.

$$H_{map \rightarrow cbv}(f)$$

MAP to CBV Transfer Function

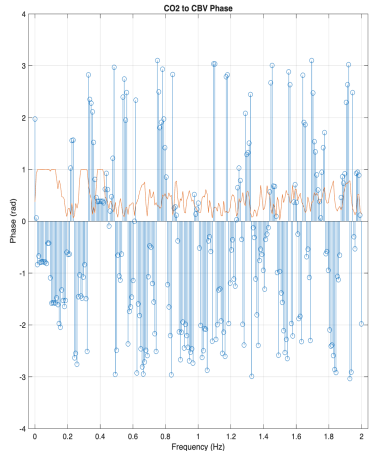
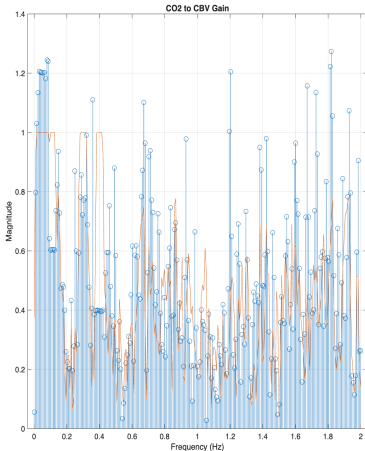


# Initial MISO Estimate: CO<sub>2</sub> to CBV

- ▶ We also estimated the CO<sub>2</sub>-to-CBV transfer function from the same MISO system.
- ▶ The orange curve shows multiple coherence, overlaid to indicate which frequencies have more reliable MISO estimates.

$$H_{co2 \rightarrow cbv}(f)$$

CO<sub>2</sub> to CBV Transfer Function

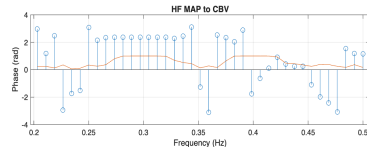
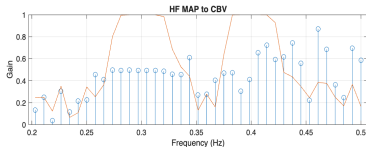
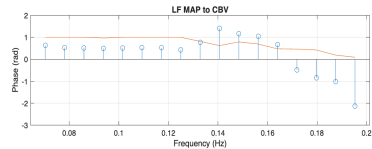
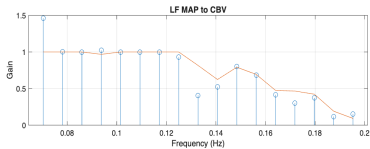
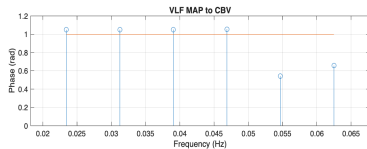
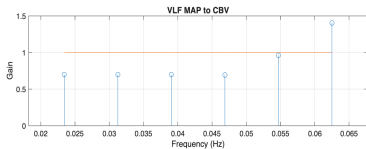


# Partitioned MISO Estimate: MAP to CBV

- ▶ Next, the MAP-to-CBV transfer function was partitioned into the standard frequency bands.
- ▶ This makes it easier to interpret gain and phase within VLF, LF, and HF ranges.

$$H_{map \rightarrow cbv}(f)$$

Partitioned MAP to CBV Transfer Function



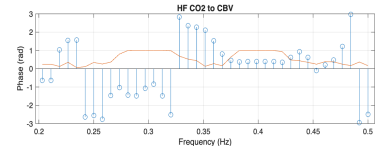
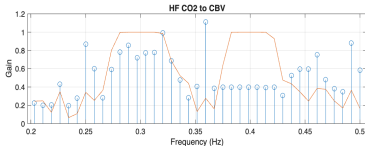
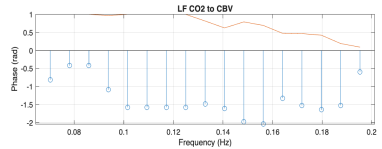
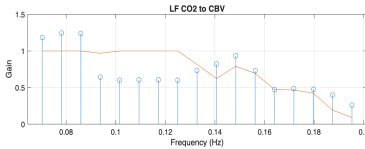
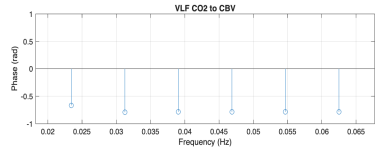
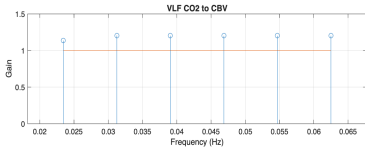


# Partitioned MISO Estimate: CO<sub>2</sub> to CBV

- ▶ The CO<sub>2</sub>-to-CBV transfer function was also partitioned into the same frequency bands.
- ▶ This allows the MAP and CO<sub>2</sub> pathways to be compared within each band.

$$H_{co2 \rightarrow cbv}(f)$$

Partitioned CO<sub>2</sub> to CBV Transfer Function

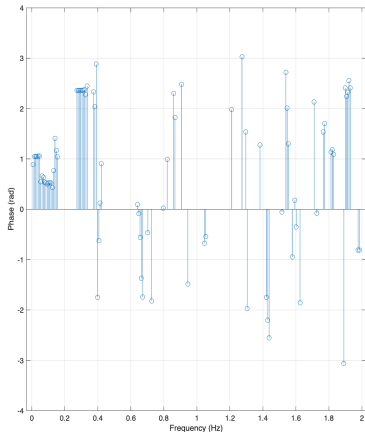
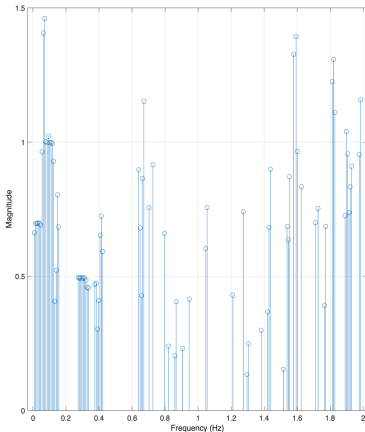


# Multiple Coherence Filtered: MAP to CBV

- ▶ Next, we filtered the MAP-to-CBV transfer function using multiple coherence.
- ▶ Multiple coherence measures how well  $map(t)$  and  $co2(t)$  together explain  $cbv(t)$  at each frequency.
- ▶ Frequencies with low multiple coherence were excluded from gain and phase interpretation.

$$H_{map \rightarrow cbv}(f)$$

Filtered MAP to CBV Transfer Function

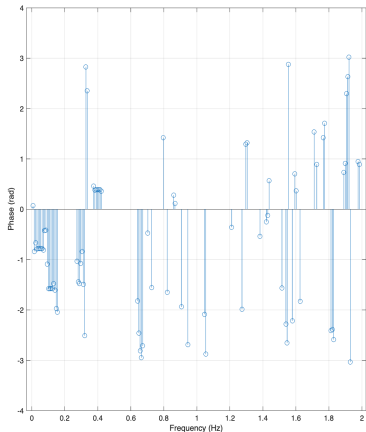
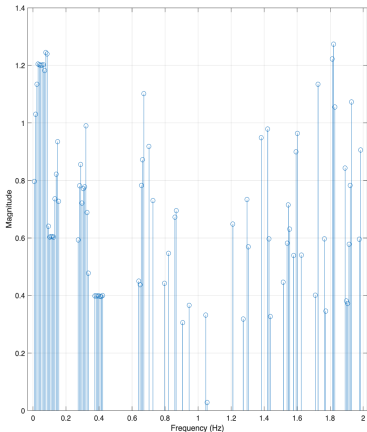


# Multiple Coherence Filtered: CO<sub>2</sub> to CBV

- ▶ The same multiple coherence filter was applied to the CO<sub>2</sub>-to-CBV transfer function.
- ▶ This keeps frequencies where the combined two-input model explains the output reliably.
- ▶ The orange curve shows the multiple coherence used for filtering.

$$H_{\text{co2} \rightarrow \text{cbv}}(f)$$

Filtered CO2 to CBV Transfer Function

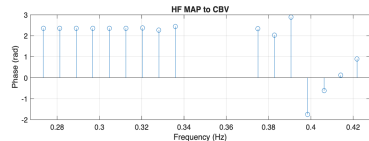
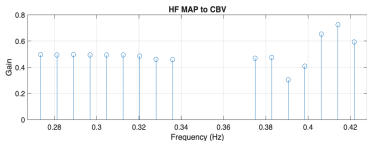
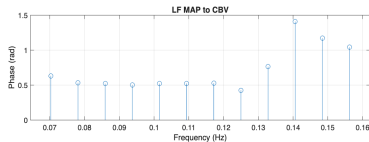
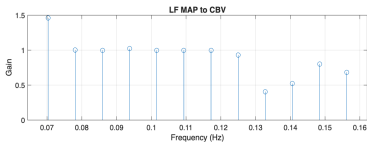
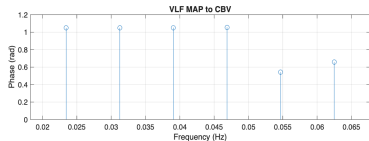
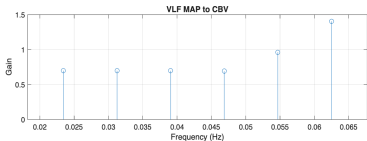


# Partitioned Multiple Coherence Filtered: MAP to CBV

- ▶ After filtering by multiple coherence, the remaining MAP-to-CBV estimates were partitioned into VLF, LF, and HF bands.
- ▶ This shows which band-specific gain and phase estimates remain after excluding low-coherence frequencies.
- ▶ The orange curve shows the multiple coherence used for filtering.

$$H_{map \rightarrow cbv}(f)$$

Filtered MAP to CBV Transfer Function

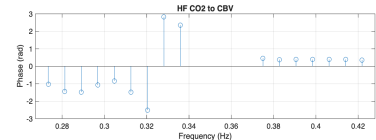
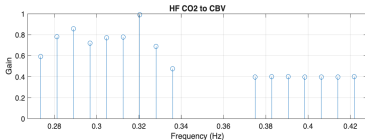
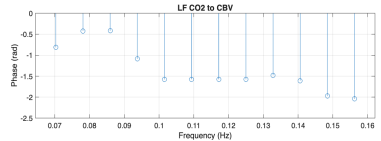
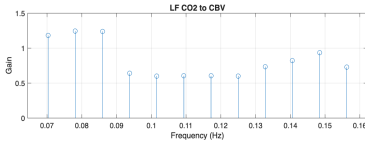
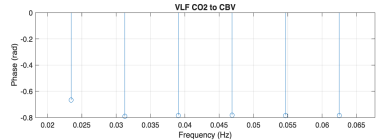
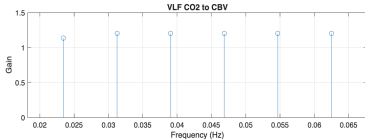


# Partitioned Multiple Coherence Filtered: CO<sub>2</sub> to CBV

- ▶ The same partitioned multiple coherence filtering was applied to the CO<sub>2</sub>-to-CBV transfer function.
- ▶ This allows the CO<sub>2</sub> pathway to be interpreted within VLF, LF, and HF after reliability filtering.
- ▶ The orange curve shows the multiple coherence used for filtering.

$$H_{co2 \rightarrow cbv}(f)$$

Filtered CO2 to CBV Transfer Function

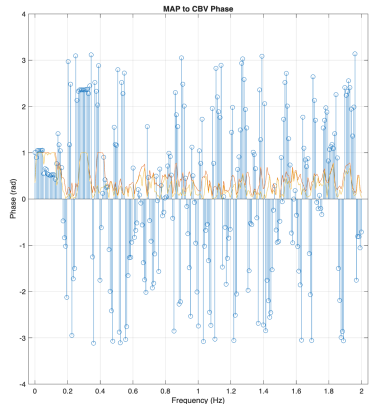
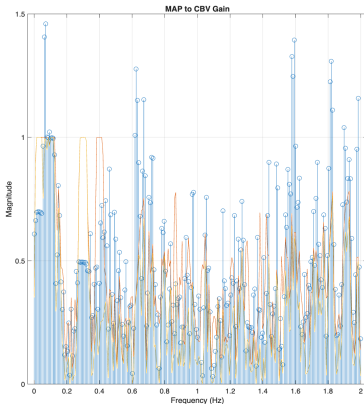


# MISO Estimate with Coherence Overlays: MAP to CBV

- ▶ The MAP-to-CBV transfer function was replotted with both coherence measures overlaid.
- ▶ The orange curve shows multiple coherence, which indicates how well  $map(t)$  and  $co2(t)$  together explain  $cbv(t)$ .
- ▶ The yellow curve shows partial coherence, which indicates how reliable the MAP-to-CBV pathway remains after accounting for  $CO_2$ .

$$H_{map \rightarrow cbv}(f)$$

MAP to CBV Transfer Function

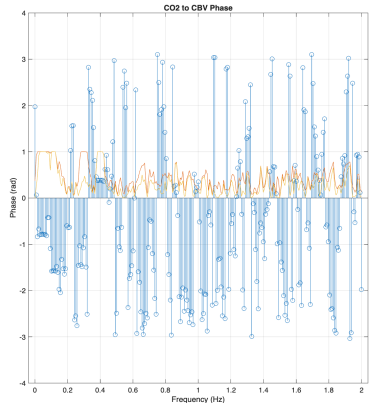
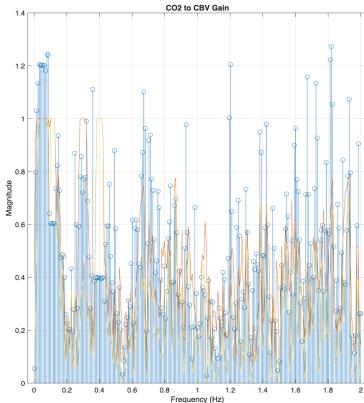


# MISO Estimate with Coherence Overlays: CO<sub>2</sub> to CBV

- ▶ The CO<sub>2</sub>-to-CBV transfer function was also replotted with both coherence measures overlaid.
- ▶ The orange curve shows multiple coherence, which indicates how well  $map(t)$  and  $co2(t)$  together explain  $cbv(t)$ .
- ▶ The yellow curve shows partial coherence, which indicates how reliable the CO<sub>2</sub>-to-CBV pathway remains after accounting for MAP.

$$H_{co2 \rightarrow cbv}(f)$$

CO2 to CBV Transfer Function

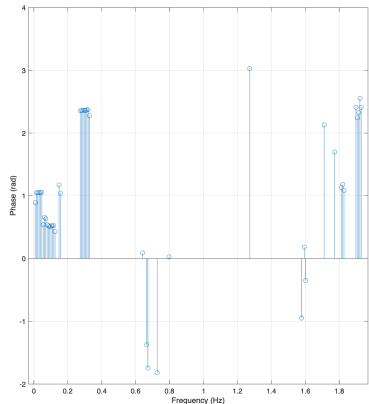
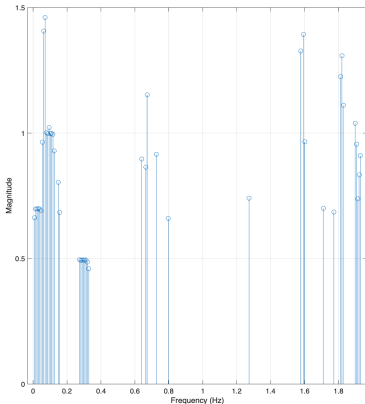


# Filtered MISO Estimate: MAP to CBV

- ▶ The MAP-to-CBV transfer function was filtered using both multiple coherence and partial coherence.
- ▶ Multiple coherence keeps frequencies where the full two-input model explains  $cbv(t)$ .
- ▶ Partial coherence keeps frequencies where the MAP-to-CBV pathway remains reliable after accounting for  $CO_2$ .

$$H_{map \rightarrow cbv}(f)$$

Multiple & Partial Coherence Filtered MAP to CBV Transfer Function



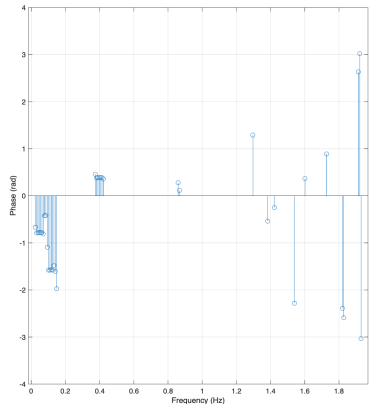
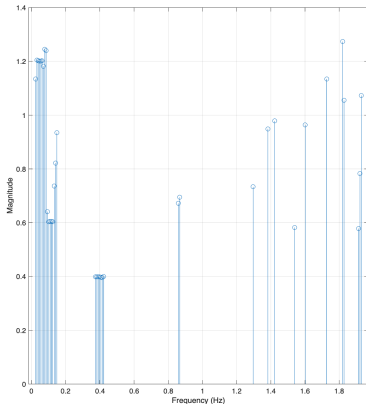


# Filtered MISO Estimate: CO<sub>2</sub> to CBV

- ▶ The CO<sub>2</sub>-to-CBV transfer function was filtered using the same combined coherence criteria.
- ▶ Multiple coherence keeps frequencies where  $map(t)$  and  $co2(t)$  together explain  $cbv(t)$ .
- ▶ Partial coherence keeps frequencies where the CO<sub>2</sub>-to-CBV pathway remains reliable after accounting for MAP.

$$H_{co2 \rightarrow cbv}(f)$$

Multiple & Partial Coherence Filtered CO2 to CBV Transfer Function

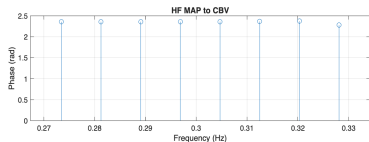
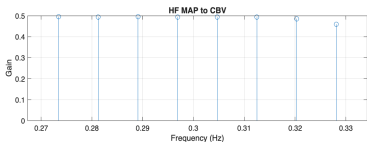
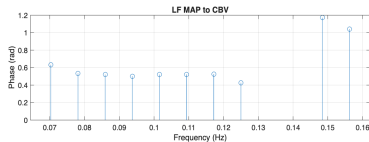
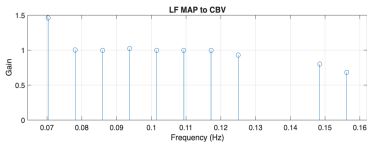
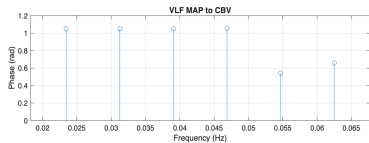
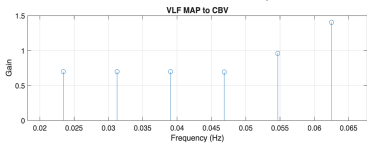


# Partitioned Filtered MISO Estimate: MAP to CBV

- ▶ The combined coherence-filtered MAP-to-CBV transfer function was then partitioned into VLF, LF, and HF bands.
- ▶ This shows the final interpretable MAP-to-CBV gain and phase estimates within each frequency range.

$$H_{map \rightarrow cbv}(f)$$

Multiple + Partial Coherence Filtered MAP to CBV Transfer Function

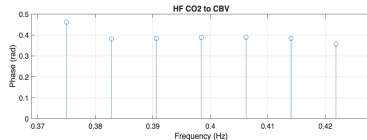
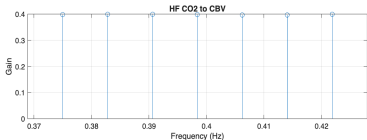
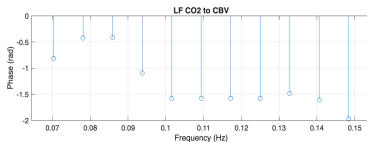
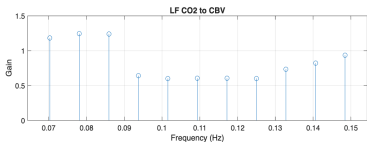
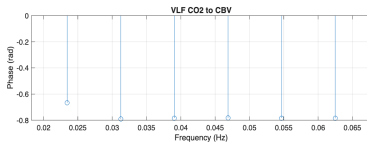
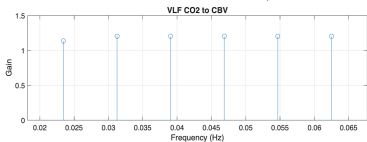


# Partitioned Filtered MISO Estimate: CO<sub>2</sub> to CBV

- ▶ The combined coherence-filtered CO<sub>2</sub>-to-CBV transfer function was also partitioned into VLF, LF, and HF bands.
- ▶ This shows the final interpretable CO<sub>2</sub>-to-CBV gain and phase estimates within each frequency range.

$$H_{co2 \rightarrow cbv}(f)$$

Multiple + Partial Coherence Filtered CO2 to CBV Transfer Function



# SISO Comparison: Why Compare Against MISO?

- ▶ To test whether MISO provides an advantage, we compared it against separate SISO estimates.
- ▶ In the SISO approach, each input-output pathway is estimated independently:

$$H_{map \rightarrow cbv}^{SISO}(f) = \frac{S_{map,cbv}(f)}{S_{map,map}(f)}$$

$$H_{co2 \rightarrow cbv}^{SISO}(f) = \frac{S_{co2,cbv}(f)}{S_{co2,co2}(f)}$$

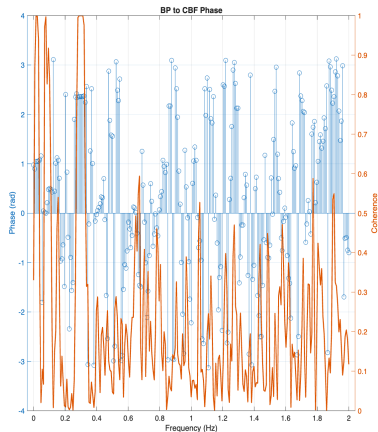
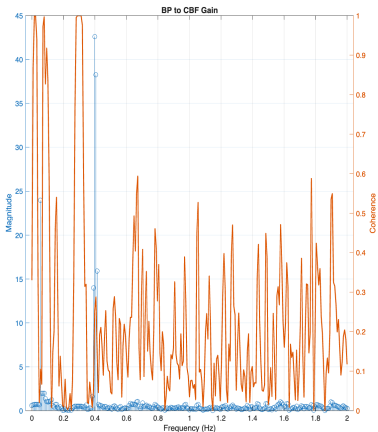
- ▶ The key test is how SISO handles shared or overlapping frequency content between  $map(t)$  and  $co2(t)$ .
- ▶ In this simulation, both inputs contain a shared LF component at 0.08 Hz.

# SISO Estimate: MAP to CBV

- ▶ The MAP-to-CBV pathway was first estimated using a separate SISO transfer function.
- ▶ This estimate only considers  $map(t)$  as the input and does not account for  $co2(t)$ .

$$H_{map \rightarrow cbv}^{SISO}(f)$$

BP to CBF Transfer Function

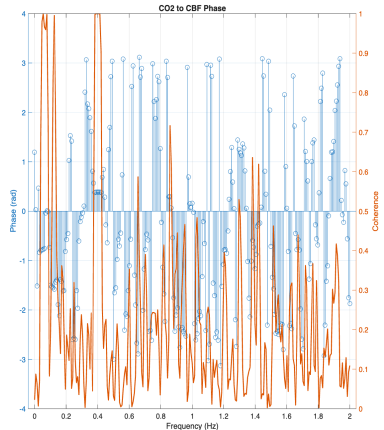
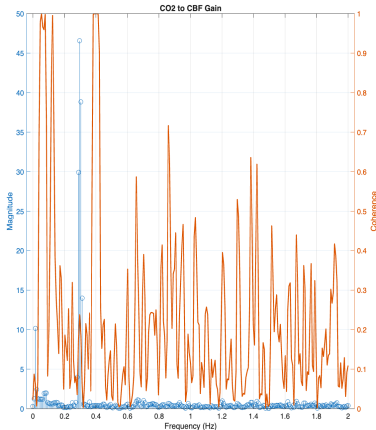


# SISO Estimate: CO<sub>2</sub> to CBV

- ▶ The CO<sub>2</sub>-to-CBV pathway was also estimated using a separate SISO transfer function.
- ▶ This estimate only considers  $co_2(t)$  as the input and does not account for  $map(t)$ .

$$H_{co_2 \rightarrow cbv}^{SISO}(f)$$

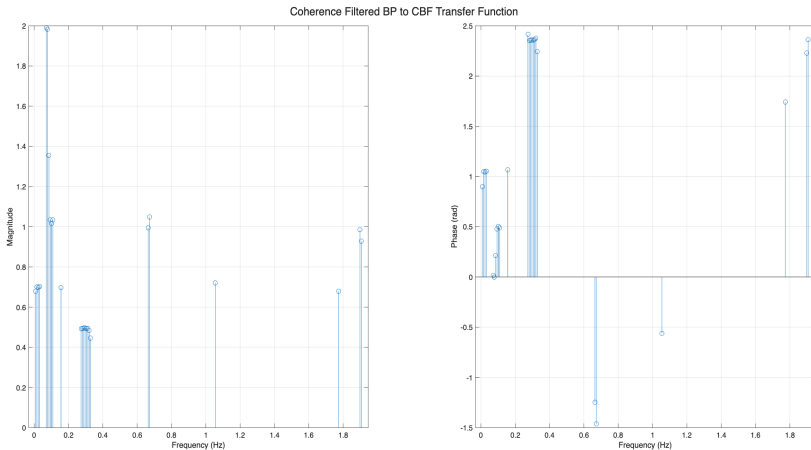
CO2 to CBF Transfer Function



# Coherence-Filtered SISO Estimate: MAP to CBV

- ▶ The SISO MAP-to-CBV estimate was filtered using univariate coherence.
- ▶ Univariate coherence only measures the reliability of the single-input relationship:

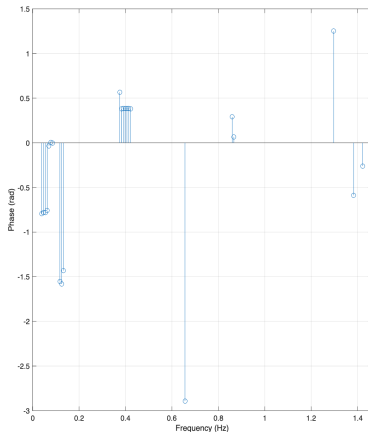
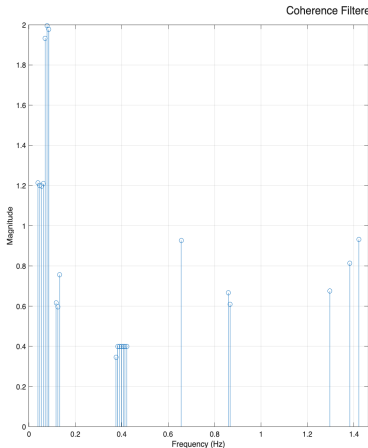
$$map(t) \rightarrow cbv(t)$$



# Coherence-Filtered SISO Estimate: CO<sub>2</sub> to CBV

- ▶ The SISO CO<sub>2</sub>-to-CBV estimate was also filtered using univariate coherence.
- ▶ This keeps frequencies where  $co2(t)$  and  $cbv(t)$  appear strongly related in the SISO model.

$$co2(t) \rightarrow cbv(t)$$



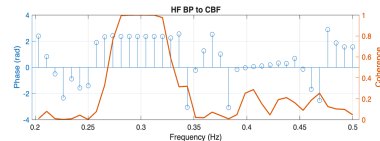
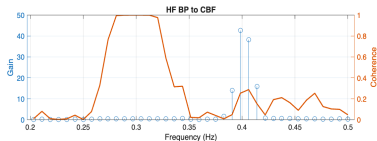
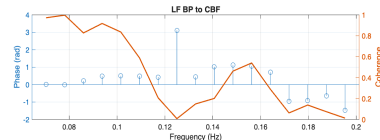
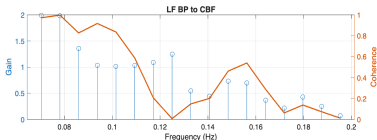
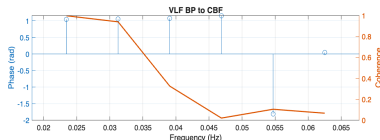
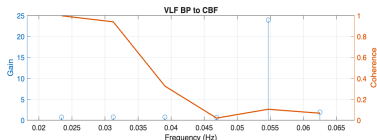


# Partitioned SISO Estimate: MAP to CBV

- ▶ The SISO MAP-to-CBV transfer function was partitioned into VLF, LF, and HF bands.
- ▶ This allows comparison with the final partitioned MISO estimates.

$$H_{map \rightarrow cbv}^{SISO}(f)$$

Partitioned BP to CBF Transfer Function

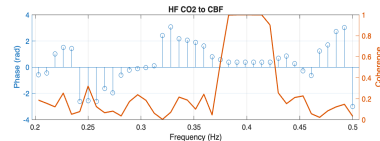
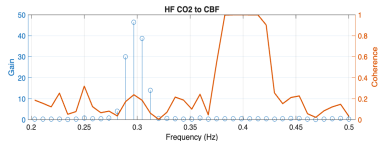
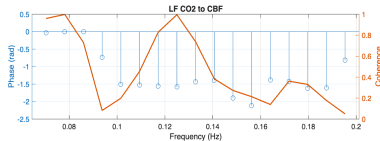
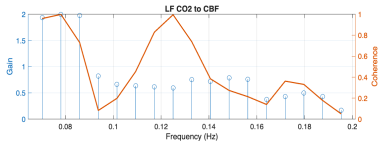
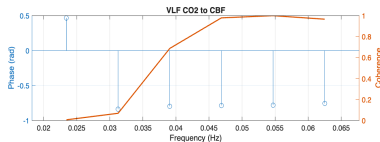
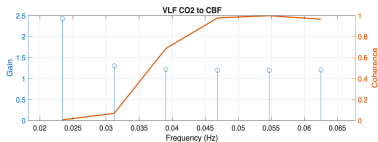


# Partitioned SISO Estimate: CO<sub>2</sub> to CBV

- ▶ The SISO CO<sub>2</sub>-to-CBV transfer function was also partitioned into VLF, LF, and HF bands.
- ▶ This helps show where SISO may over-attribute shared frequency content.

$$H_{co2 \rightarrow cbv}^{SISO}(f)$$

Partitioned CO2 to CBF Transfer Function

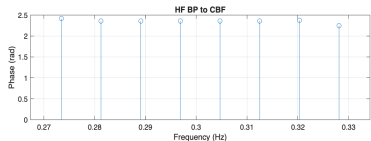
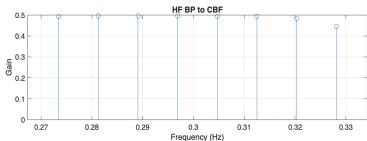
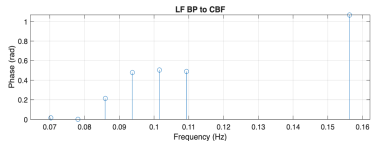
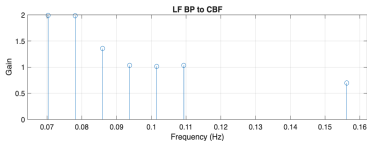
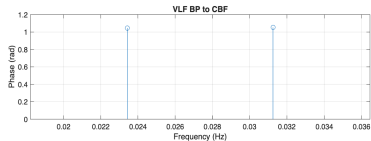
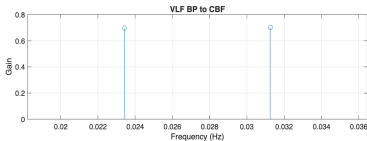


# Partitioned Coherence-Filtered SISO: MAP to CBV

- ▶ After applying univariate coherence filtering, the remaining MAP-to-CBV SISO estimates were partitioned into VLF, LF, and HF bands.
- ▶ This gives the band-specific SISO estimates after excluding low-coherence frequencies.

$$H_{map \rightarrow cbv}^{SISO}(f)$$

Coherence Filtered BP to CBF Transfer Function

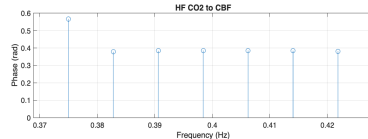
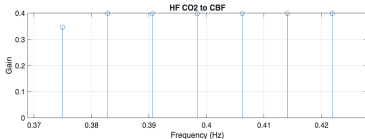
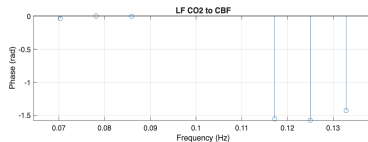
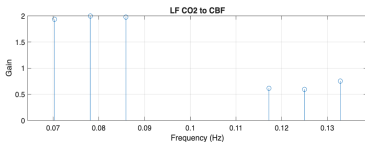
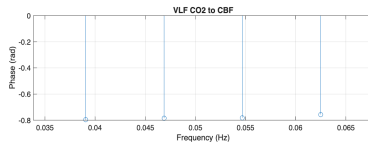
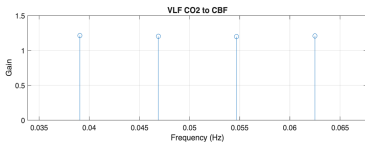


# Partitioned Coherence-Filtered SISO: CO<sub>2</sub> to CBV

- ▶ The coherence-filtered CO<sub>2</sub>-to-CBV SISO estimates were also partitioned into VLF, LF, and HF bands.
- ▶ These plots are useful for comparing against the final partitioned MISO estimates.

$$H_{co2 \rightarrow cbv}^{SISO}(f)$$

Coherence Filtered CO2 to CBF Transfer Function



# Shared Frequency: Why SISO Double-Counts

- ▶ The simulation intentionally included a shared LF component at 0.08 Hz.
- ▶ Both inputs contained the same shared component:

$$map_{shared}(t) = 0.4 \sin(2\pi(0.08)t)$$

$$co2_{shared}(t) = 0.4 \sin(2\pi(0.08)t)$$

- ▶ The output contained a shared-frequency component with amplitude 0.8:

$$cbv_{shared}(t) = 0.8 \sin(2\pi(0.08)t)$$

- ▶ In SISO, each pathway is estimated independently, so each input can attribute the full shared CBV response to itself:

$$H^{SISO}(0.08) \approx \frac{0.8}{0.4} = 2$$

- ▶ This is why both SISO pathways can show inflated gain near 2 at the shared frequency.

# MISO: Accounting for Shared Input Content

- ▶ MISO estimates both pathways simultaneously:

$$CBV(f) = H_{map \rightarrow cbv}(f)MAP(f) + H_{co2 \rightarrow cbv}(f)CO2(f)$$

- ▶ At the shared 0.08 Hz frequency:

$$0.8 = 0.4H_{map \rightarrow cbv}(0.08) + 0.4H_{co2 \rightarrow cbv}(0.08)$$

- ▶ Unlike SISO, MISO represents the shared output as a combined contribution from both inputs.
- ▶ In this simulation, the MISO estimates were closer to:

$$H_{map \rightarrow cbv}(0.08) \approx 1 \quad H_{co2 \rightarrow cbv}(0.08) \approx 1.2$$

- ▶ This is more reasonable than the SISO result of approximately 2 for both pathways, because the shared CBV response is no longer assigned entirely to each input separately.

# Important Nuance: The Shared Split Is Not Unique

- ▶ The shared-frequency equation is:

$$0.8 = 0.4H_{map} + 0.4H_{co2}$$

$$H_{map} + H_{co2} = 2$$

- ▶ This equation alone does not uniquely determine the individual gains.
- ▶ For example, all of these could explain the same shared output:

$$H_{map} = 1, \quad H_{co2} = 1$$

$$H_{map} = 1.5, \quad H_{co2} = 0.5$$

$$H_{map} = 0.2, \quad H_{co2} = 1.8$$

- ▶ Therefore, MISO avoids obvious SISO double-counting, but the pathway-specific split is only reliable when the inputs contain enough independent information.

# What Does Input Overlap Mean?

- ▶ Input overlap means MAP and CO<sub>2</sub> contain related frequency content.
- ▶ **Perfect overlap:** same frequency with a fixed amplitude/phase relationship.

$$CO_2(f) = ae^{j\phi} MAP(f)$$

- ▶ In this case, the inputs carry nearly the same information at that frequency, so the pathway split is difficult or impossible to identify uniquely.
- ▶ **Near overlap:** frequencies are close but not identical, such as 0.10 Hz and 0.11 Hz.
- ▶ These can be separated if the data length/window gives enough frequency resolution.
- ▶ **Partial overlap:** inputs share some frequency content but also have independent components.
- ▶ This is where MISO is useful: it accounts for shared input content while estimating both pathways simultaneously.



# Input-Input Coherence as an Overlap Warning

- ▶ Input-input coherence measures how similar MAP and CO<sub>2</sub> are at each frequency:

$$\gamma_{map,co2}^2(f) = \frac{|S_{map,co2}(f)|^2}{S_{map,map}(f)S_{co2,co2}(f)}$$

- ▶ Low input-input coherence means the inputs have more independent information.
- ▶ High input-input coherence means the inputs are strongly overlapping.
- ▶ High input-input coherence does not mean the full MISO model is wrong, but it warns that individual pathway gains may be harder to separate.

# Why High Input-Input Coherence Creates Uncertainty

- ▶ The MISO transfer functions are solved from:

$$G_{xx}(f)H(f) = G_{xy}(f)$$

$$\begin{bmatrix} S_{map,map} & S_{map,co2} \\ S_{co2,map} & S_{co2,co2} \end{bmatrix} \begin{bmatrix} H_{map \rightarrow cbv} \\ H_{co2 \rightarrow cbv} \end{bmatrix} = \begin{bmatrix} S_{map,cbv} \\ S_{co2,cbv} \end{bmatrix}$$

- ▶ Solving requires inverting  $G_{xx}(f)$ . For two inputs, the inverse contains the denominator:

$$S_{map,map}S_{co2,co2} - S_{map,co2}S_{co2,map}$$

- ▶ Since  $S_{co2,map} = S_{map,co2}^*$ :

$$S_{map,map}S_{co2,co2} - |S_{map,co2}|^2$$

# Input Coherence Appears in the Matrix Inverse

- ▶ From input-input coherence:

$$\gamma_{map,co2}^2(f) = \frac{|S_{map,co2}(f)|^2}{S_{map,map}(f)S_{co2,co2}(f)}$$

- ▶ Rearranging:

$$|S_{map,co2}(f)|^2 = \gamma_{map,co2}^2(f)S_{map,map}(f)S_{co2,co2}(f)$$

- ▶ Substituting this into the inverse denominator gives:

$$\begin{aligned} & S_{map,map}S_{co2,co2} - |S_{map,co2}|^2 \\ &= S_{map,map}S_{co2,co2} (1 - \gamma_{map,co2}^2) \end{aligned}$$

- ▶ Therefore, as input-input coherence approaches 1, the denominator of the matrix inverse approaches 0.

# Interpreting High Input-Input Coherence

- ▶ If  $map(t)$  and  $co2(t)$  are not strongly overlapping:

$$\gamma_{map,co2}^2(f) \text{ is low} \quad \Rightarrow \quad 1 - \gamma_{map,co2}^2(f) \text{ is not close to } 0$$

- ▶ The matrix inversion is more stable, and the pathway-specific gains are more identifiable.
- ▶ But if the inputs are highly overlapping:

$$\gamma_{map,co2}^2(f) \rightarrow 1 \quad \Rightarrow \quad 1 - \gamma_{map,co2}^2(f) \rightarrow 0$$

- ▶ Then small spectral estimation errors can strongly affect the estimated transfer functions.
- ▶ The computer can still output a number, but the pathway-specific gain may be unstable or ambiguous.
- ▶ Therefore, input-input coherence is not an output reliability measure; it is an identifiability warning.

# Takeaway: MISO Helps, But Reliability Still Matters

- ▶ SISO can double-count shared frequency content because each pathway is estimated independently.
- ▶ MISO accounts for MAP–CO<sub>2</sub> overlap through the off-diagonal terms of  $G_{xx}(f)$ :

$$G_{xx}(f) = \begin{bmatrix} S_{map,map}(f) & S_{map,co2}(f) \\ S_{co2,map}(f) & S_{co2,co2}(f) \end{bmatrix}$$

- ▶ However, if MAP and CO<sub>2</sub> are too coherent with each other, the exact pathway split becomes less reliable.
- ▶ Multiple coherence tells whether the full MISO model explains  $cbv(t)$ .
- ▶ Partial coherence tells whether each individual pathway remains interpretable after accounting for the other input.
- ▶ Input-input coherence warns whether the inputs are too overlapping to separate cleanly.

# Next Steps and Future Directions

- ▶ **Apply to real physiological data**
  - ▶ Test the PSD-MISO pipeline on real  $map(t)$ ,  $co2(t)$ , and  $cbv(t)$  recordings.
  - ▶ Compare SISO vs. MISO gain, phase, and coherence across VLF, LF, and HF bands.
- ▶ **Evaluate reliability and identifiability**
  - ▶ Use multiple coherence to assess whether the full two-input model explains  $cbv(t)$ .
  - ▶ Use partial coherence to determine whether each pathway remains interpretable.
  - ▶ Use input-input coherence to flag frequencies where MAP and  $CO_2$  overlap too strongly.
- ▶ **Develop a wavelet-based MISO approach**
  - ▶ Extend  $H(f)$  to a time-varying transfer function  $H(t, f)$ .
  - ▶ Use wavelet coherence, multiple wavelet coherence, and partial wavelet coherence to study nonstationary autoregulation.